

FILTER DESIGN TEST CASE

Multiplexer Module - Band 66 (AWS) for Automotive V2X Communication
Design Validation and Performance Characterization

DOCUMENT INFORMATION

Document Number:	FD-TC-0082
Revision:	Rev C.1
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Created Date:	2025-03-13
Last Modified:	2025-04-16
Status:	NEEDS REVIEW
Priority:	P3 - Medium

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Multiplexer Module filter targeting Band 66 (AWS) for Automotive V2X Communication. The filter is designed on AlN on Si substrate with a target center frequency of 3602.0 MHz and bandwidth of 169.7 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Automotive V2X Communication module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Multiplexer Module
- Frequency Band: Band 66 (AWS)
- Application: Automotive V2X Communication
- Substrate: AlN on Si
- Package: Flip-Chip BGA
- Design Tool: PathWave EM Design
- Design Center: Newbury Park Design Center

This specification applies to all variants of the Multiplexer Module design targeting Band 66 (AWS). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3602.0 MHz
Bandwidth (3 dB):	169.7 MHz
Insertion Loss Target:	<= 2.9 dB
Return Loss Target:	>= 17.0 dB
Out-of-Band Rejection:	>= 23.0 dB
Isolation (Duplexer):	>= 30.0 dB
Q Factor:	12780
Temperature Coefficient:	-14.4 ppm/°C
Die Size:	0.8 x 0.9 mm
Wafer Size:	8-inch
Number of Resonators:	10
Electrode Material:	Mo

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open PathWave EM Design and load the Multiplexer Module template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 3602.0 MHz, BW 169.7 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.9 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, AlN on Si).
10. Perform on-wafer probing using Rohde & Schwarz ZNA67.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 10806 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.9 dB
- Return loss in passband shall be >= 17.0 dB
- Out-of-band rejection at +/- 255 MHz offset >= 23.0 dB
- Group delay variation across passband shall be <= 3 ns
- Temperature drift over -40°C to +85°C shall be <= 6.5 MHz
- Power handling shall be >= +27 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 96%

6. TEST RESULTS

Measurement Date: 2025-04-16
Devices Tested: 83
Insertion Loss (meas): 2.62 dB
Return Loss (meas): 18.6 dB
Rejection (meas): 25.8 dB
Isolation (meas): 34.9 dB
Q Factor (meas): 12780
Group Delay Variation: 5.1 ns
Power Handling: +35 dBm
Die Yield: 89.8%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

- a) The Multiplexer Module design demonstrated below-target performance for Band 66 (AWS).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Rejection at near-band offset requires additional resonator tuning.
- d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
- e) No delamination observed after 1000-cycle thermal shock test.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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